

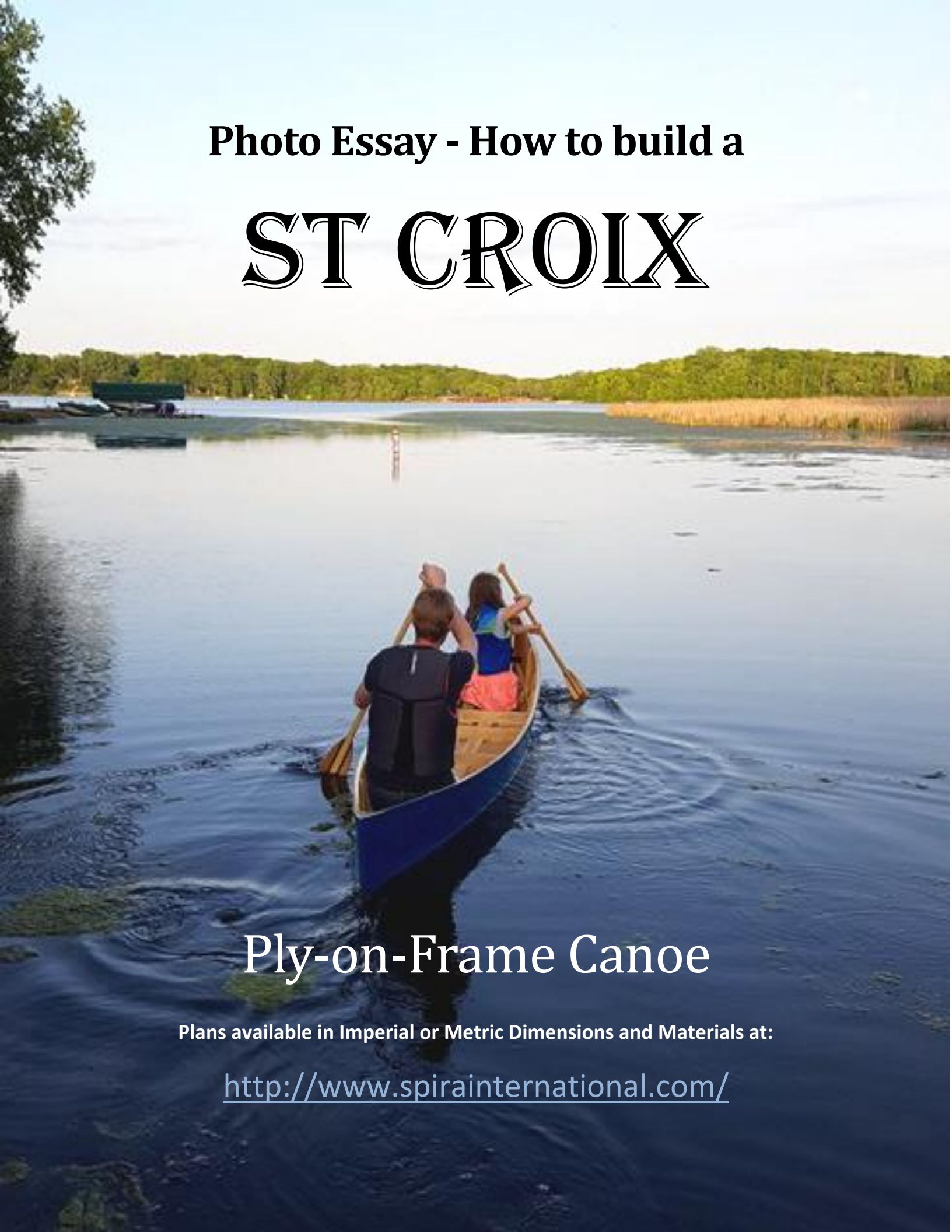
A photograph of a man and a woman paddling a blue canoe on a calm lake. The man is in the front, wearing a black shirt and a life vest, and the woman is in the back, wearing a blue shirt and a pink skirt. They are both using wooden paddles. The lake is surrounded by a dense forest of green trees. In the background, there are some wooden structures and a small boat. The sky is clear and blue.

Photo Essay - How to build a

ST CROIX

Ply-on-Frame Canoe

Plans available in Imperial or Metric Dimensions and Materials at:

<http://www.spirainternational.com/>

How to Build the St Croix Ply on Frame Canoe

By Jeff Spira

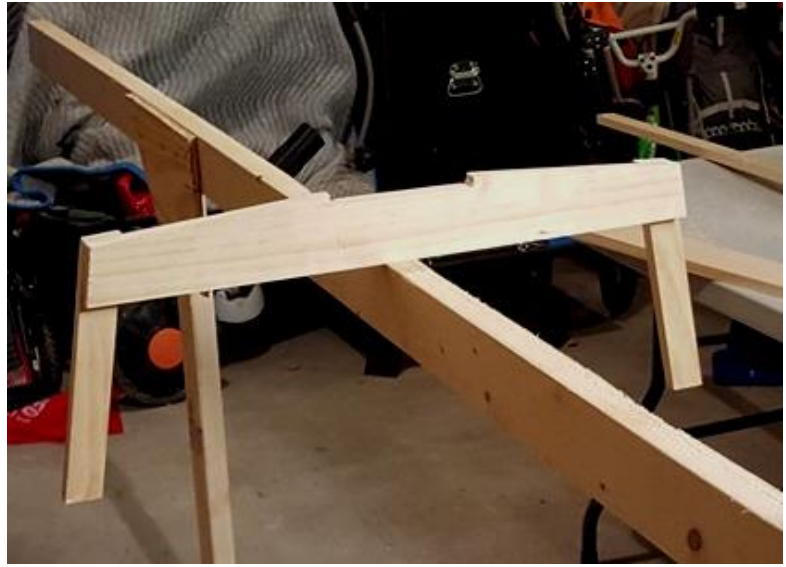
The St. Croix is a traditional canoe. It's built of the popular, ply on frame construction method so can be built with modern materials and methods by the home builder. It is a V-Bottom design so is extra slippery and extra east to build. No sophisticated techniques or methods are required.

I named the St. Croix near where I took my first canoe ride. It is the river that separates Minnesota and Wisconsin. I spent a lot of time in the Stillwater, Minnesota area on the river and nearby lakes with my father and our friends when I lived in the St. Paul area from about 4 years old until I was 11.



The St Croix, like most Spira International, easy-to-build boats is built on a simple strongback jig. The plans how to build this are included in the boat plans. It can be made from any available construction grade lumber.

The frames are next. These are simply laid out on a work surface, cut from standard dimensional lumber, and glued and screwed together. All frame elements are straight in this boat and the bottom frame is cut to produce the v-bottom when required.



The frames are notched to receive the keelson timber a little wider than needed to create limber holes for bilge water to flow between, and then set up on the strongback blocked up to prescribed heights above the strongback.

The keelson is then fitted to the frames. There is no significant bending required. It is simply attached in place to the frames.



Next, the frames are notched for the Chine Log (along the intersection of the bottom and sides, and the Sheer Clamp (along the top edge of the frames.) These longitudinal frame members are then bent, glued and fastened into place.

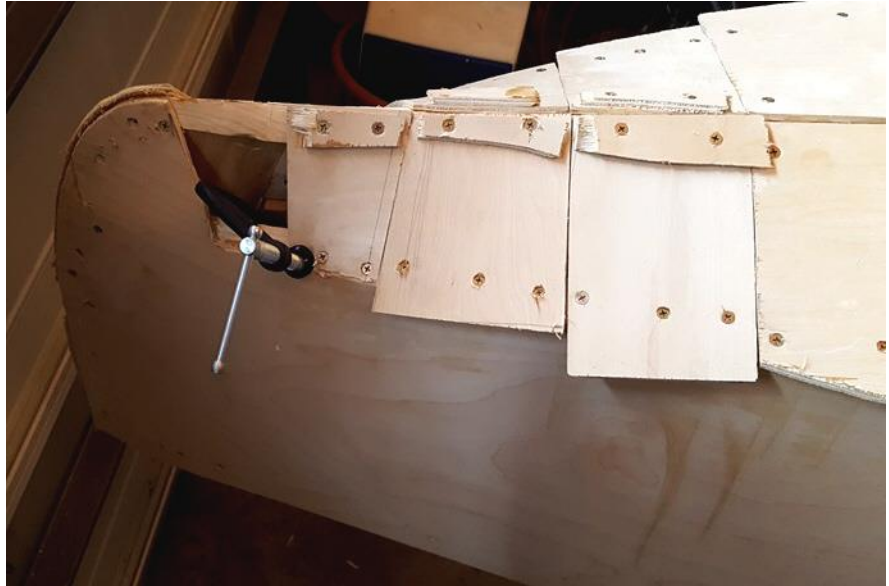
The process of “Fairing” the hull comes next. This is done with a saw, plane, drawknife, angle grinder and/or other tools to smooth the outside surface of the frames and angle then so that the ply covering will lay against the framing with the correct angle so that a good, wide glue joint can be established.



Once the hull is faired, you can begin the plywood planking process. The frame itself is used as a template

to cut out the plywood. You simply butt together the ply then later back it up where there is no frame element with a backing plate of plywood in the inside, glued and screwed in-place.

Near then ends when the ply has to twist, smaller strips can be cut out and attached to form the v-entry and v shape in the stern.



After you complete the ply covering, fill the cracks, holes and screw head holes with an epoxy/filler

mixture. You can use microspheres or silica, but it's just as easy and a lot cheaper to just use sawdust mixed with epoxy resin to make a sort of hardening putty.

Once all the holes are filled, it's time to sand the hull fully. This gives the epoxy a slightly roughened surface to stick to.



Now it is time to cut out the fiberglass to cover the hull. I recommend two layers of 6 oz. fiberglass cloth. The cloth comes 50" wide and it is sold by the yard (180 g/m³, 1270 mm width, by the meter length in metric dimensions)

You saturate the cloth with epoxy resin by first painting the bare wood with epoxy, then laying the cloth on the wet wood, and applying epoxy to the cloth to fully wet it out. It turns from opaque when dry to transparent when properly saturated. You can lay the second coat over the first while it is still wet and then wet it out as well.



All glassed.
Now it's time
to sand, sand,
and sand. Not
an easy task
but unless you
want your
boat looking
like a rough
burlap sack, a
necessary
one.



When you're done it should look something like this.



While it's upside down is a good time to paint the hull. After that is done, it's time to flip the boat over.

What do you know, it actually is starting to look like a boat.



Finish off the insides, then hoist it to the top of the car and head out to get it wet.



Ah, yes, canoeing time...

